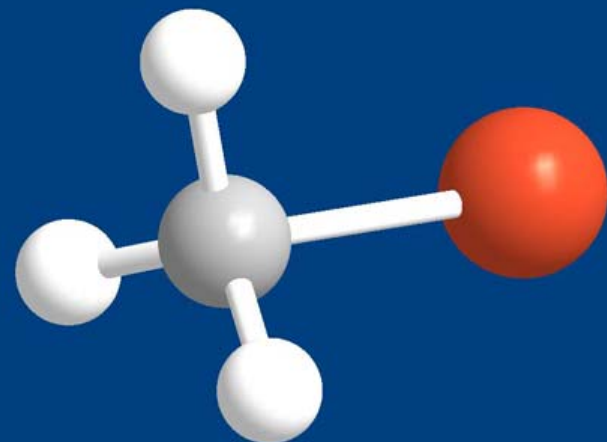




第八章 卤代烃

Alkyl Halides

厦门大学有机化学B

CONTENT

- 1 卤代烃的分类、结构和命名
- 2 卤代烃的物理性质
- 3 卤代烃的化学性质
- 4 亲核取代的反应机理
- 5 消除反应的反应机理
- 6 卤代烃的制备

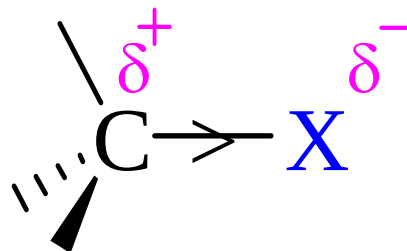
8.1 卤代烃的分类、结构和命名

8.1.1 卤代烃的分类

- 按烃基结构分类：饱和卤代烃、不饱和卤代烃、芳香卤代烃
- 按卤原子个数分类：单卤代烃、二卤代烃、多卤代烃
- 按卤素所连碳原子的结构分类：一级卤代烃、二级卤代烃、三级卤代烃

8.1.2 卤代烃的结构

- 键长、键能



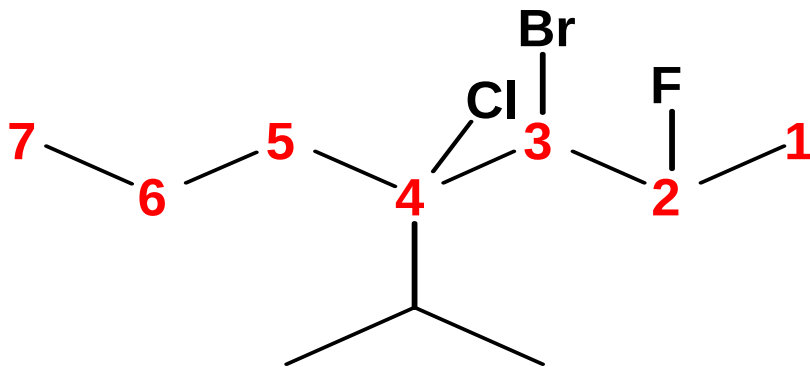
	C—F	C—Cl	C—Br	C—I
键长	139	176	194	214 pm
键能	456	351	293	243 kJ / mol

- 偶极矩

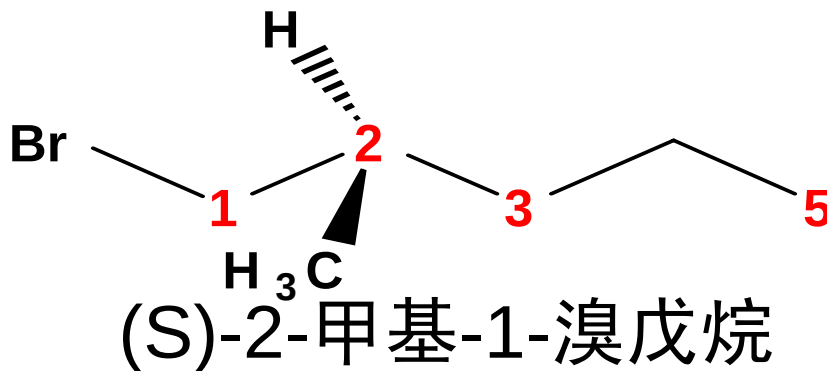
卤代烷	CH ₃ CH ₂ Cl	CH ₃ CH ₂ Br	CH ₃ CH ₂ I
偶极矩	2.05 D	2.03 D	1.91 D

8.1.3 卤代烃的命名

- 卤代烃不作为一类母体化合物
- 卤原子在命名时始终作为取代基



4-异丙基-2-氟-4-氯-3-溴庚烷



(S)-2-甲基-1-溴戊烷

8.2 卤代烃的物理性质

- **沸点：** 分子的极性越大，偶极—偶极相互作用也大，沸点升高
- **溶解度：** 卤代烃不溶于水
- **密度：** 只有一氯代烃、一氟代烃比水轻

8.3 卤代烃的化学性质

8.3.1 亲核取代反应 (S_N)

Nucleophilic Substitution Reactions



Alkyl halide

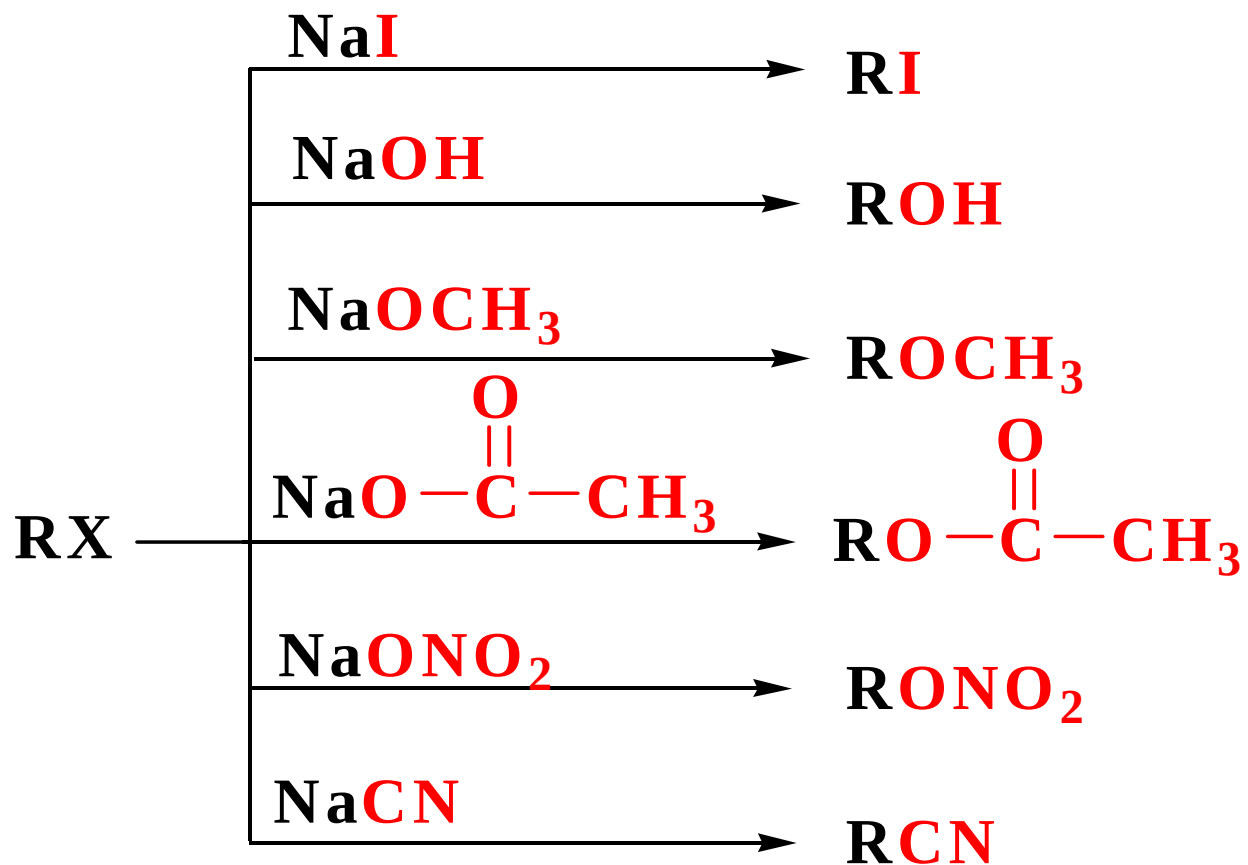
Nucleophile

Product

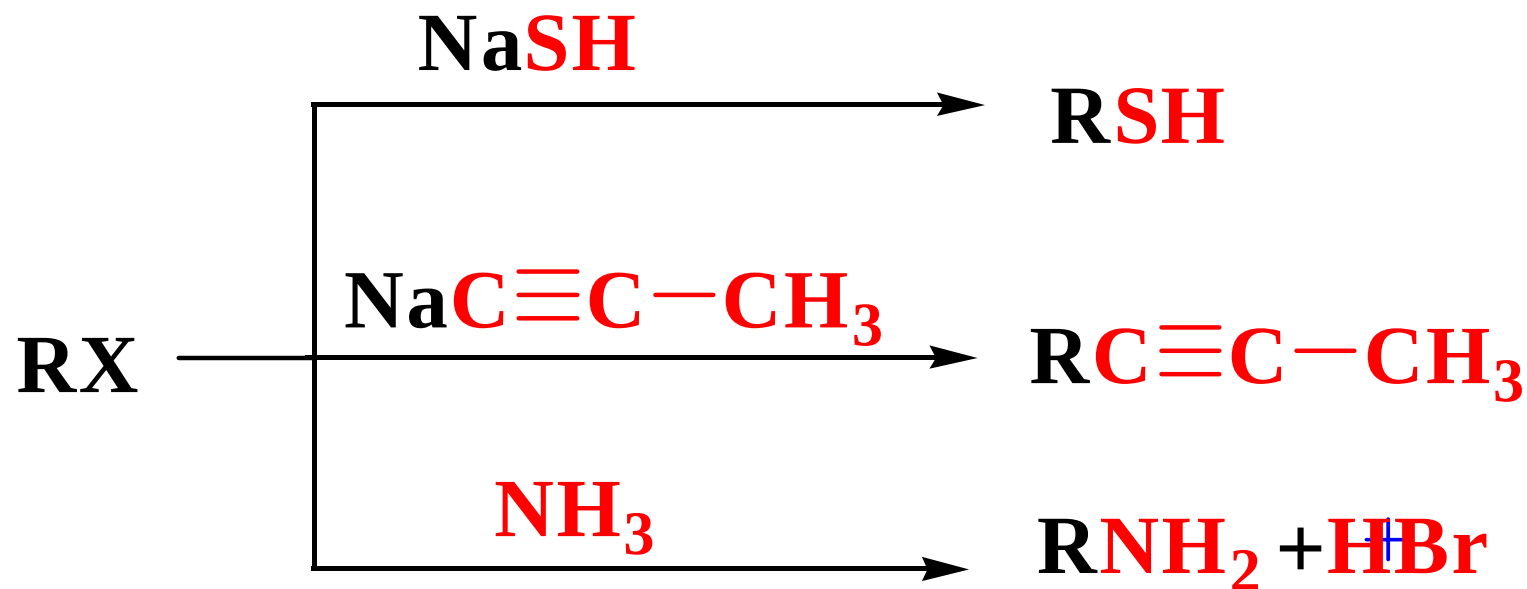
Halide ion

(底物)

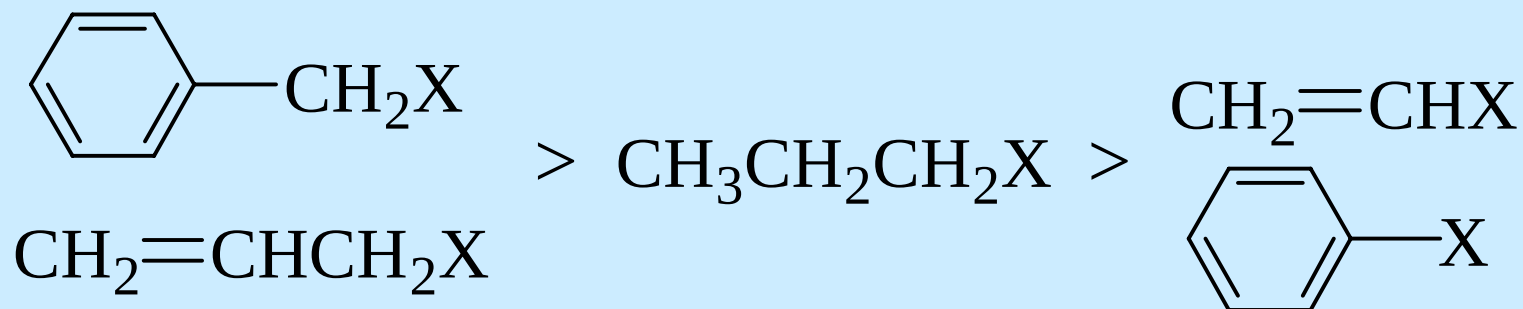
Example



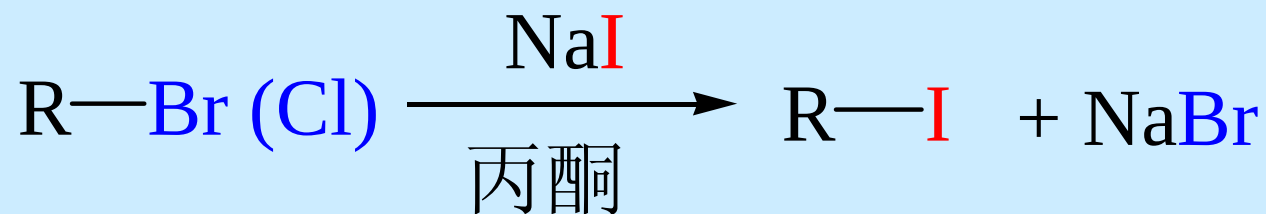
Example



卤代烃的活泼性

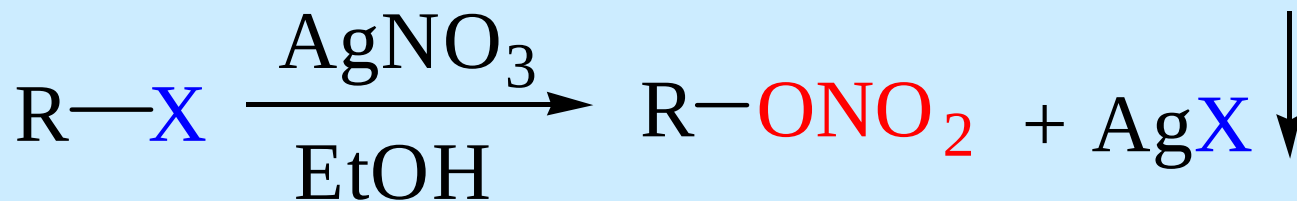


卤素交换反应



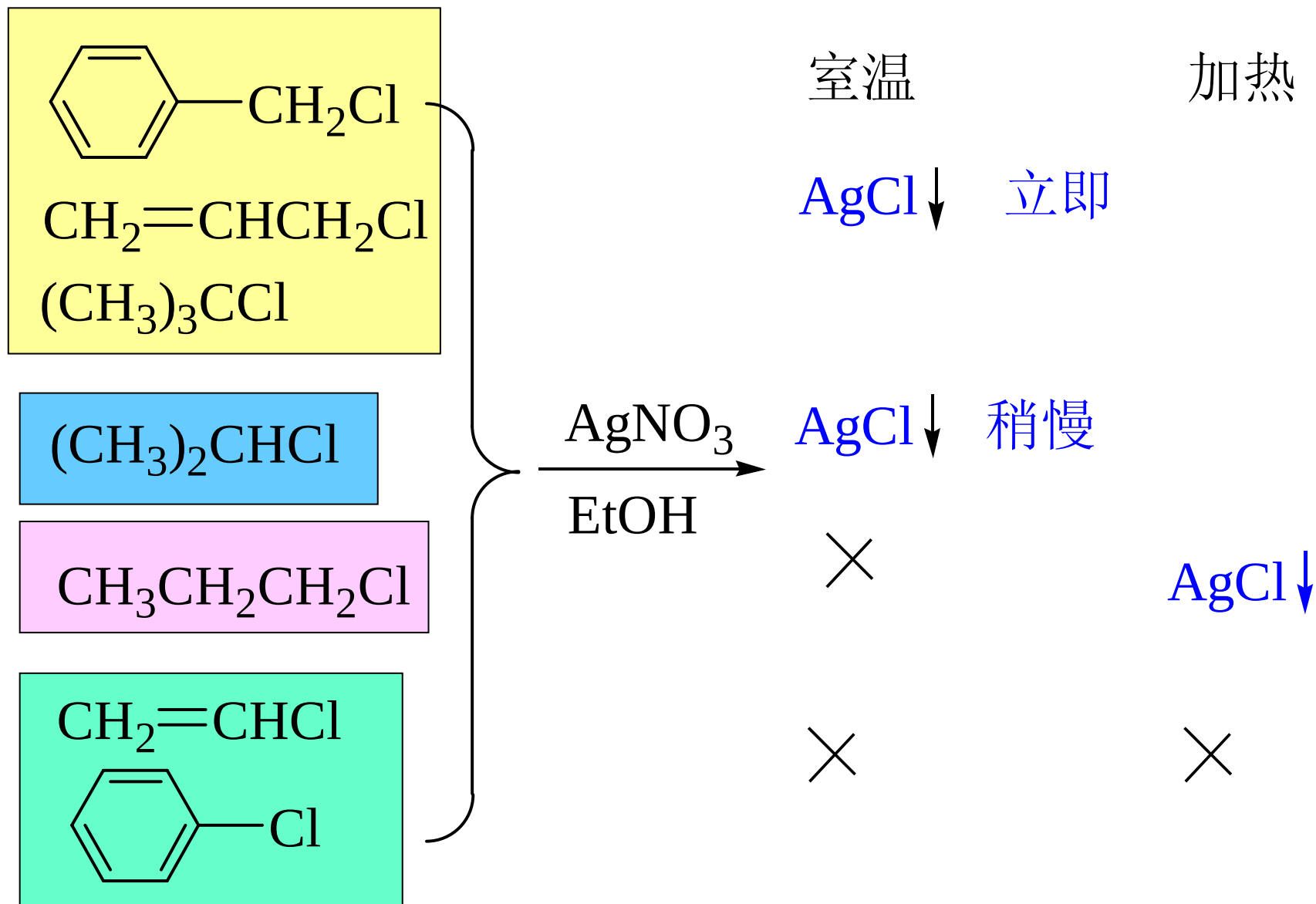
鉴定氯代烃和溴代烃

卤代烃与硝酸银的反应



用于鉴别伯、仲、叔卤代烃

Example



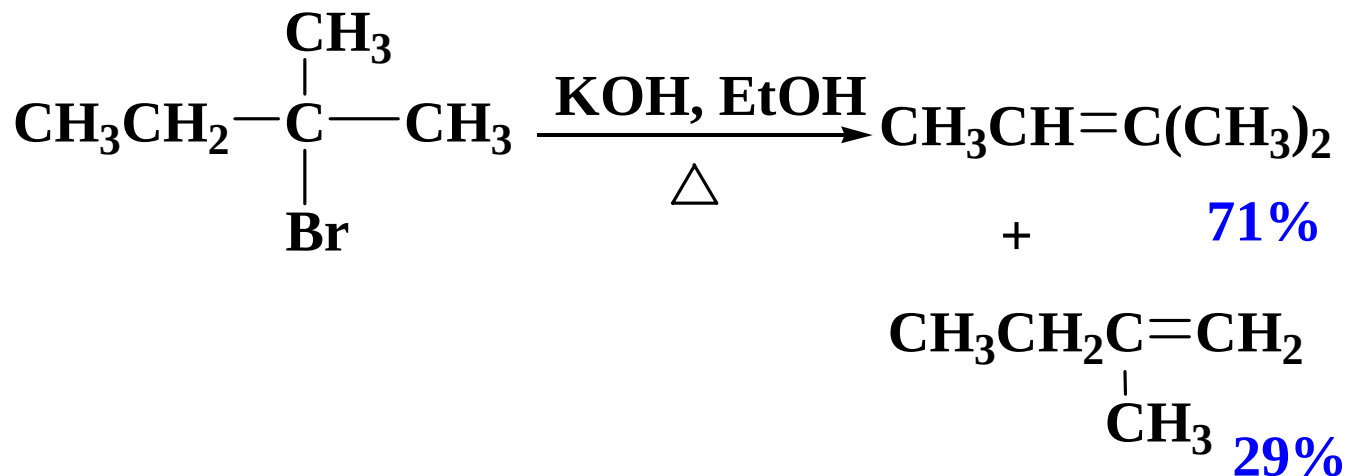
8.3.2 消除反应

Elimination Reaction

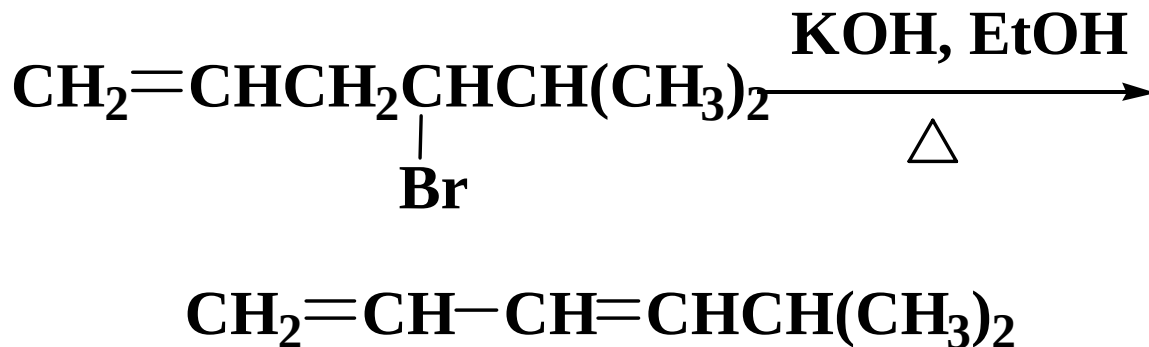
1. 脱卤化氢



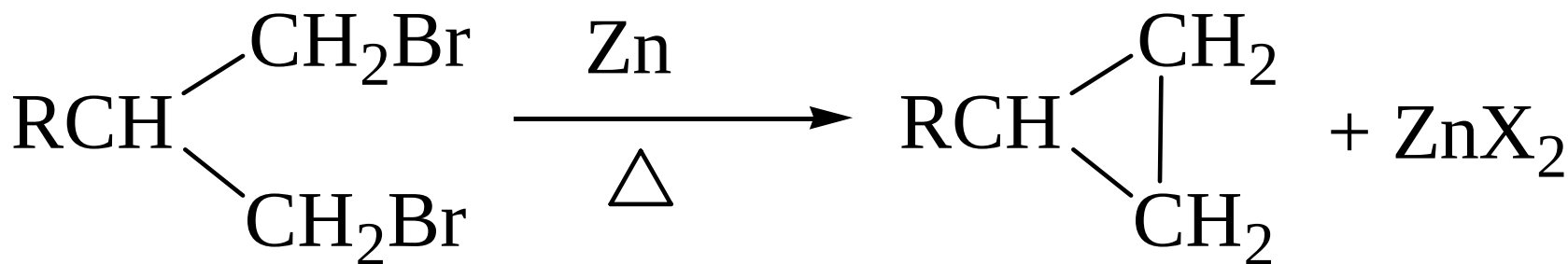
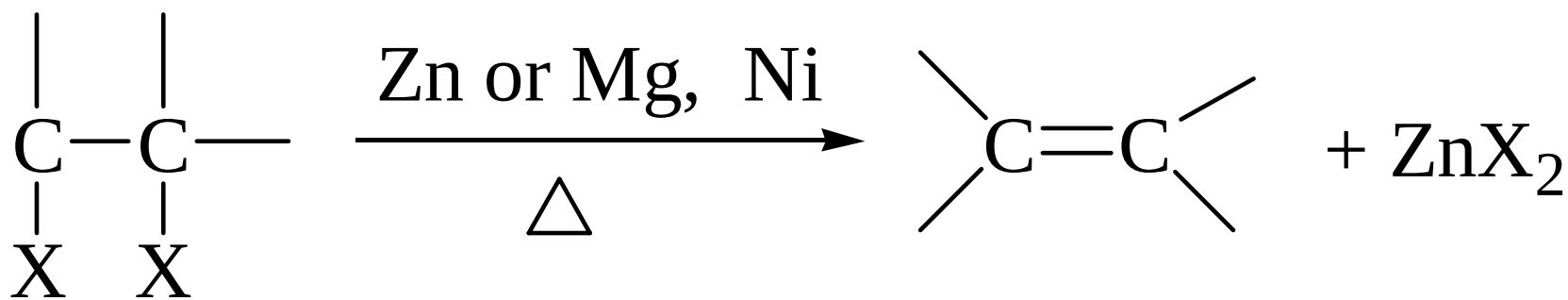
- 遵从Zaitsev (Saytzeff)规律



- 优先形成共轭烯烃



2. 脱卤素



8.3.3 还原反应



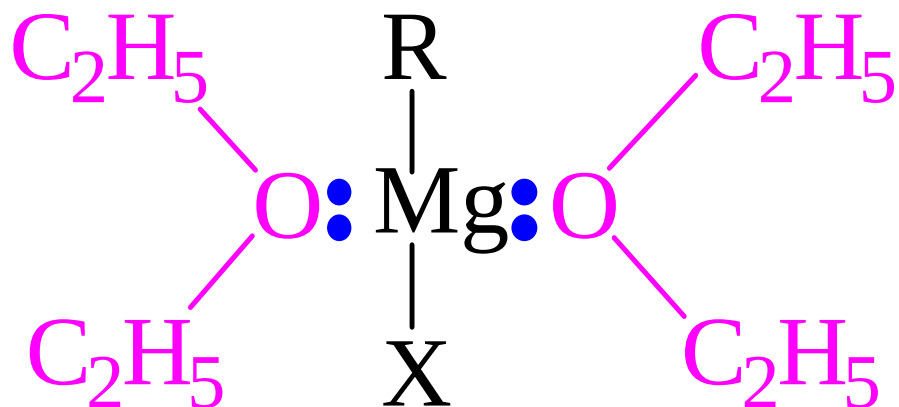
(1) 催化氢化: H_2 , Pd, Ni....

(2) 化学还原: LiAlH_4 , NaBH_4 , Zn / HCl

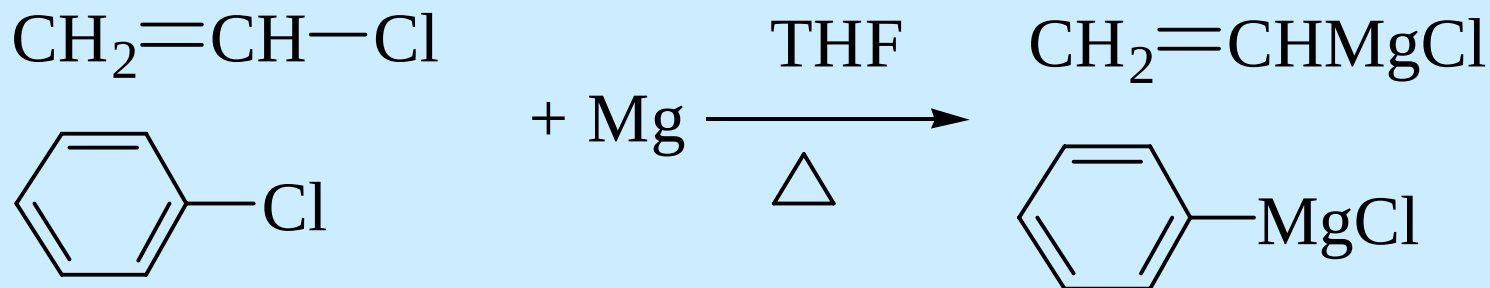
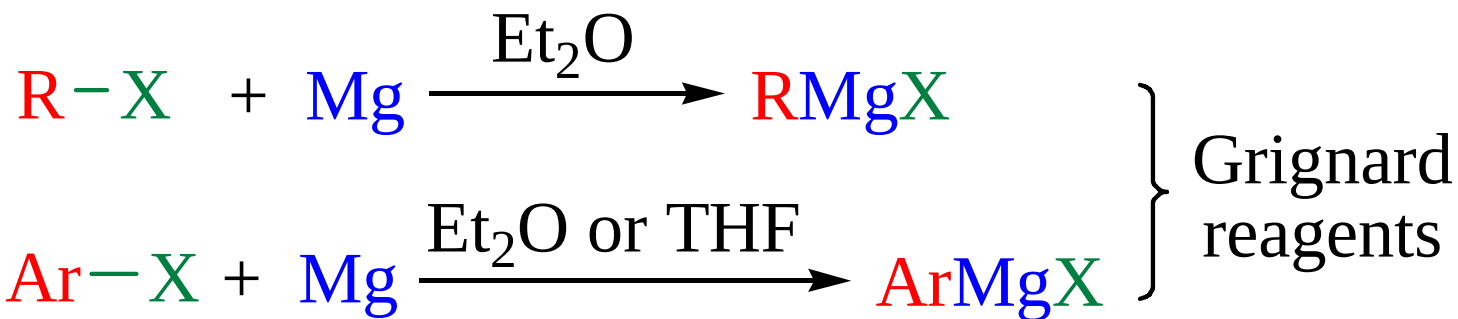
8.3.4 与金属的反应



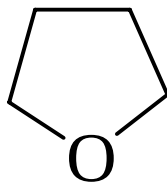
Grignard 试剂



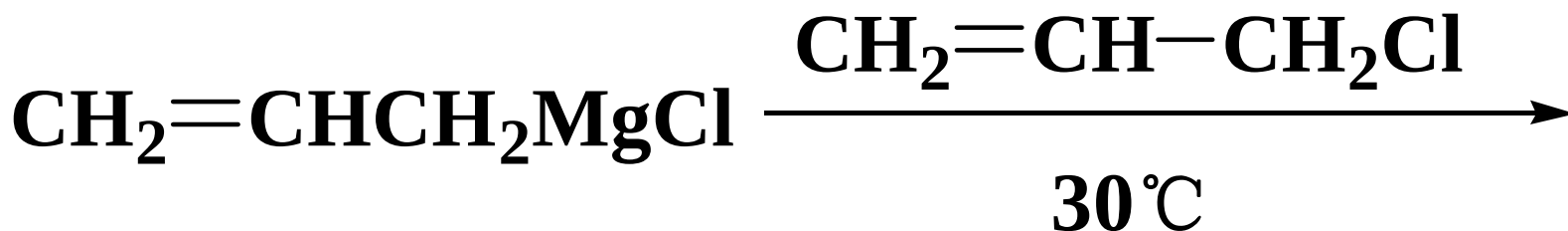
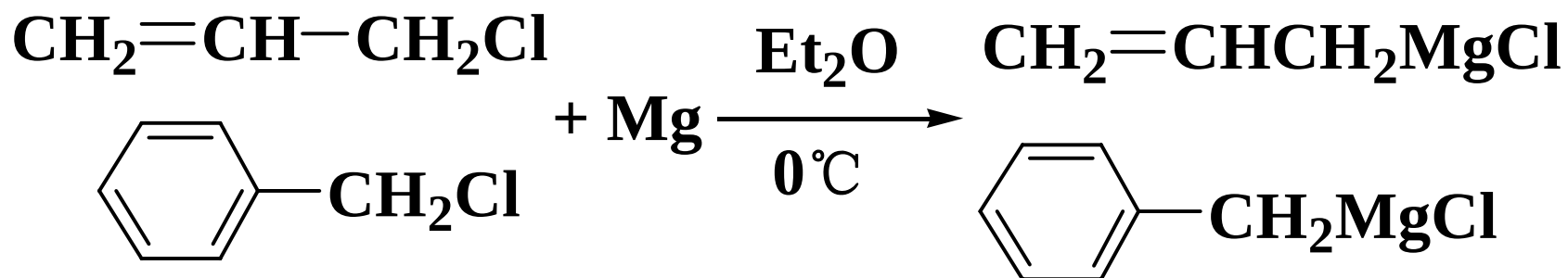
乙烯型和芳香卤代烃不活泼，须提高温度



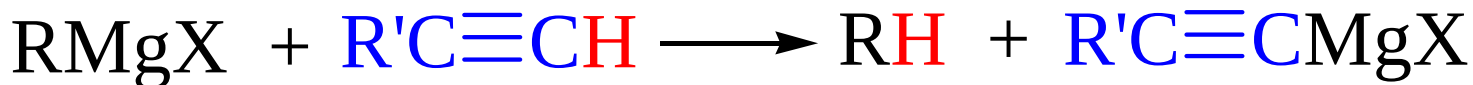
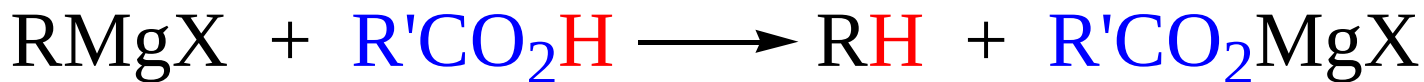
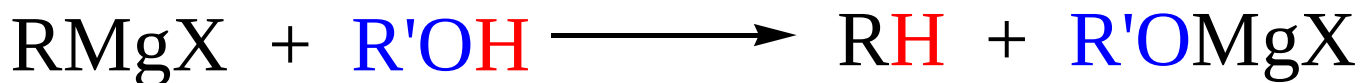
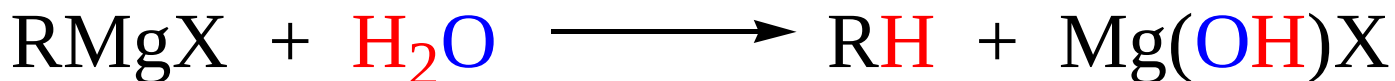
THF Tetrahydrofuran



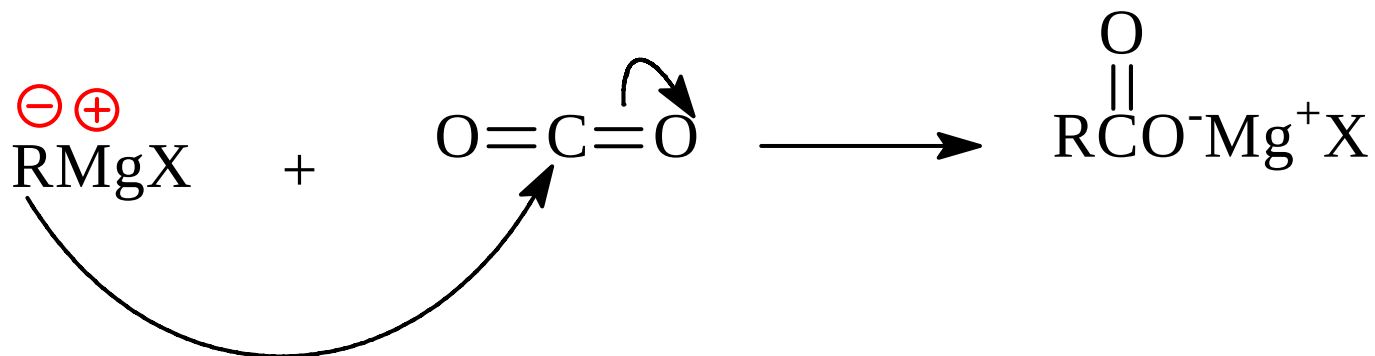
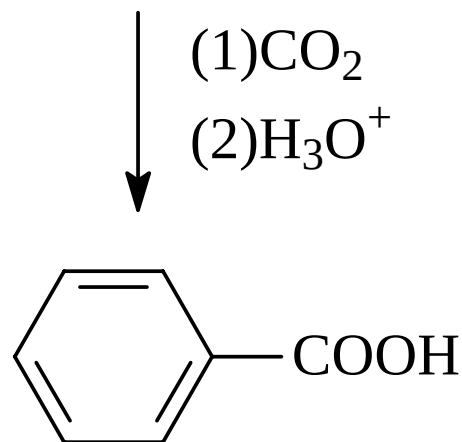
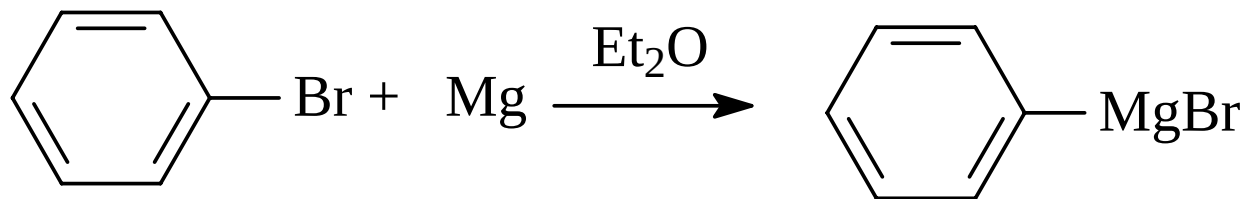
烯丙型和苄基型较活泼，要低温反应



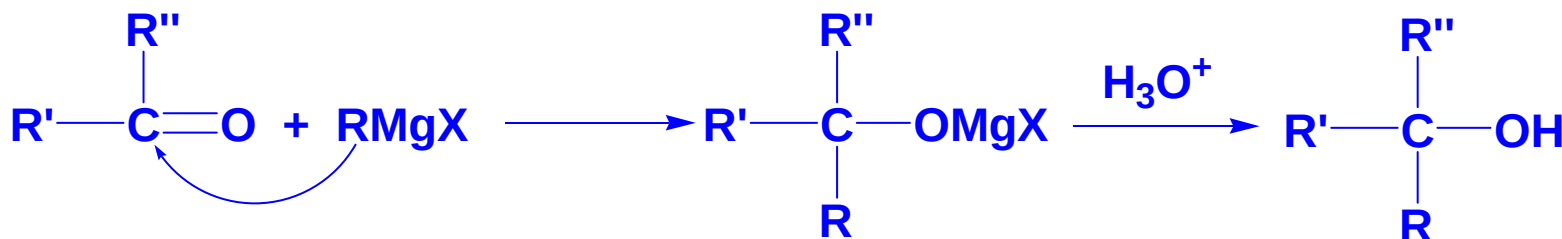
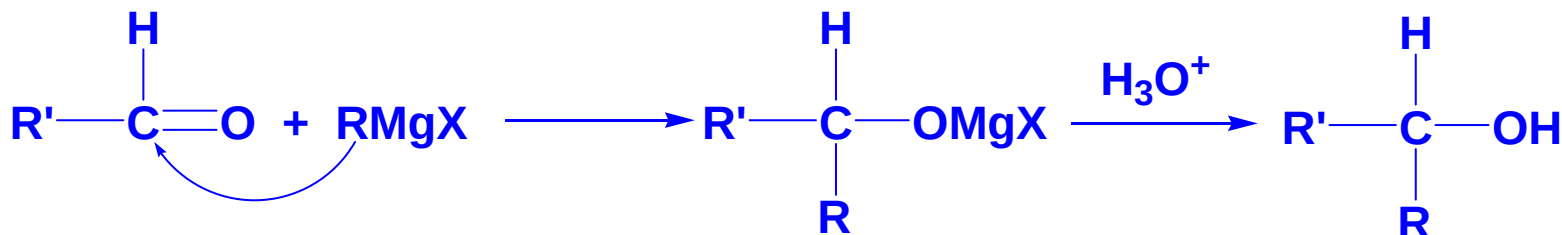
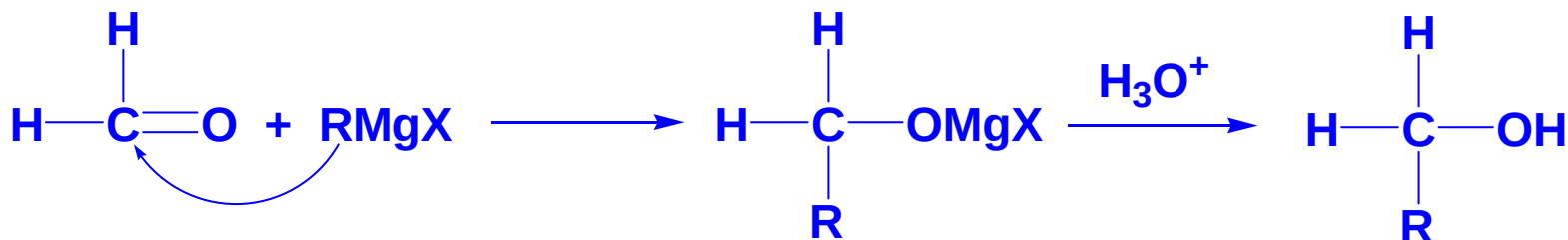
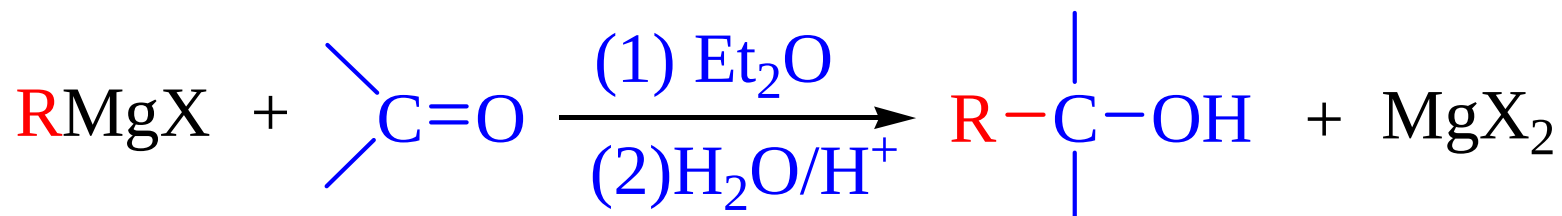
Grignard 试剂与活泼氢的反应

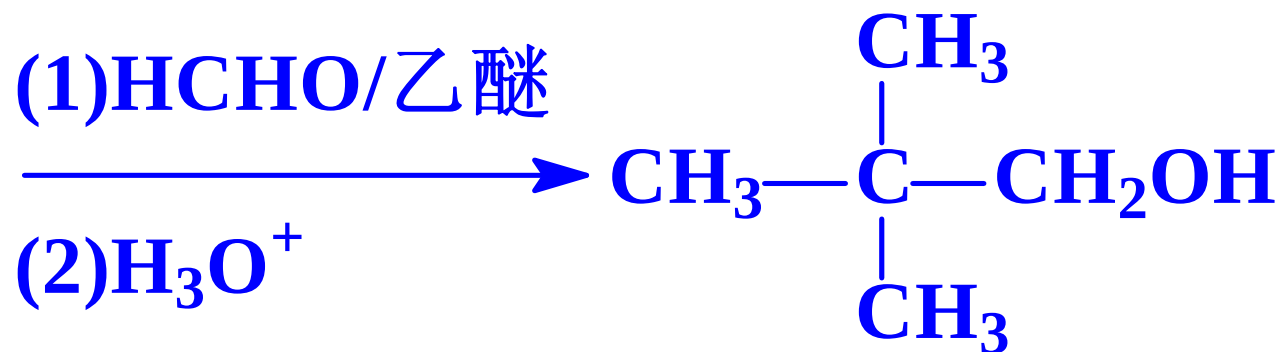
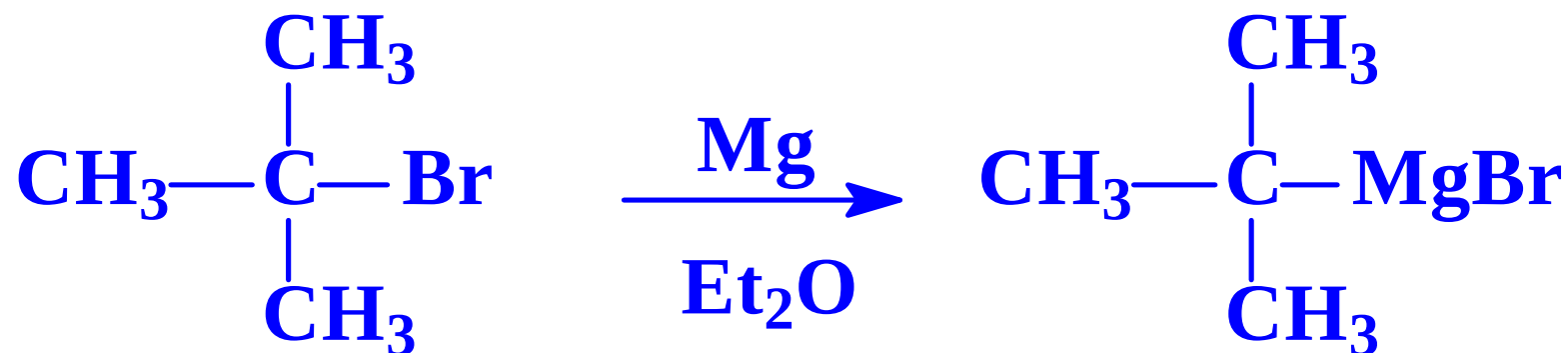


Grignard 试剂与二氧化碳反应

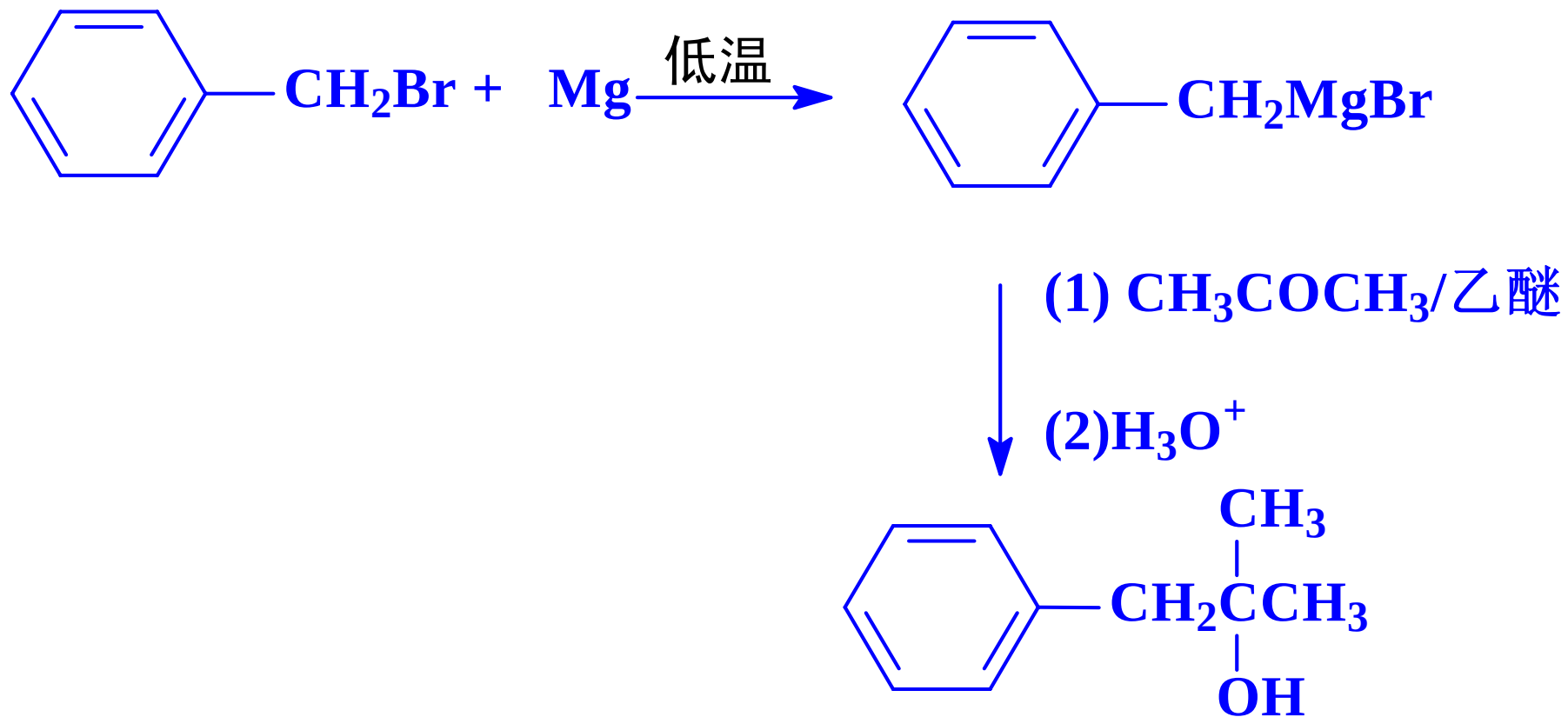


Grignard 试剂与醛酮反应

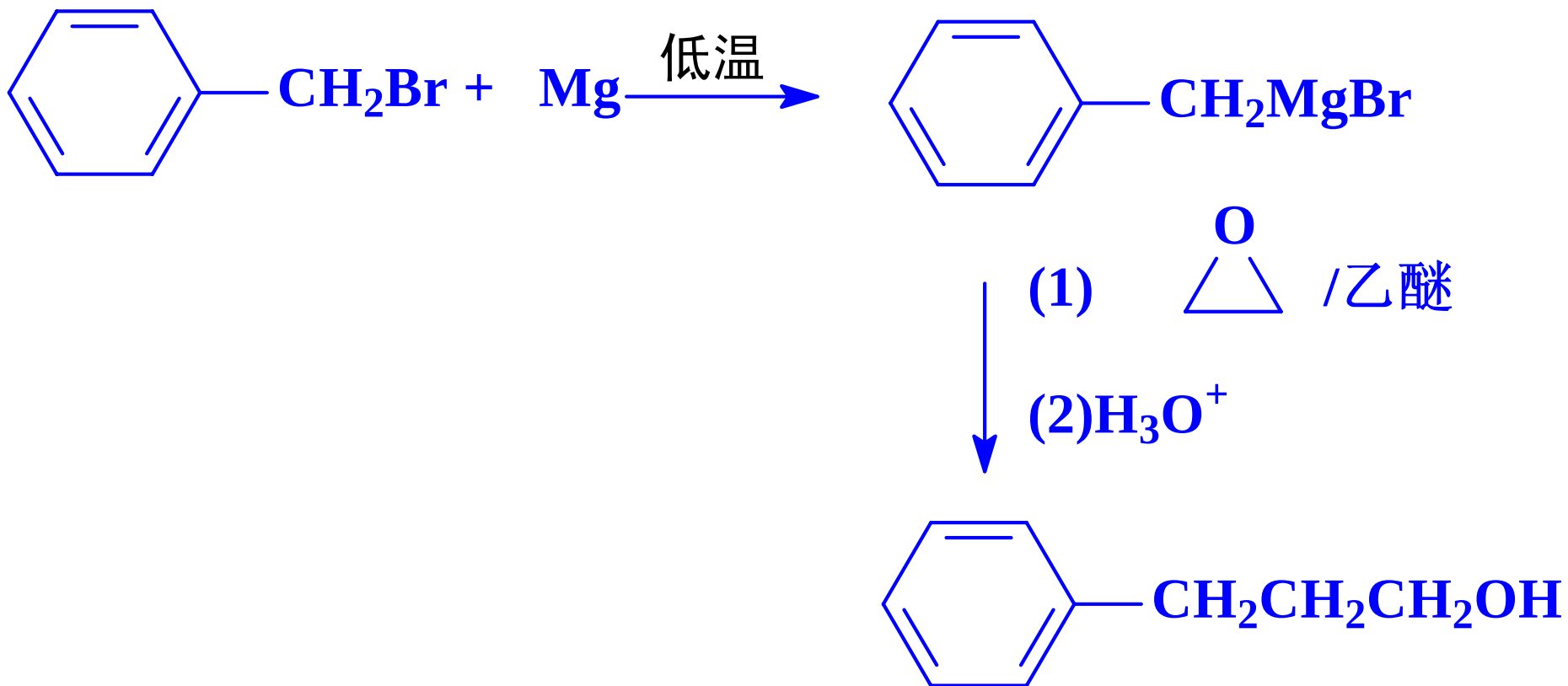




Example



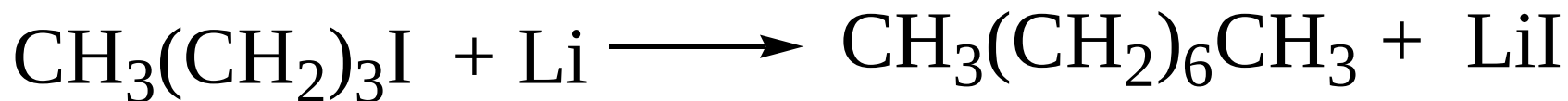
Example



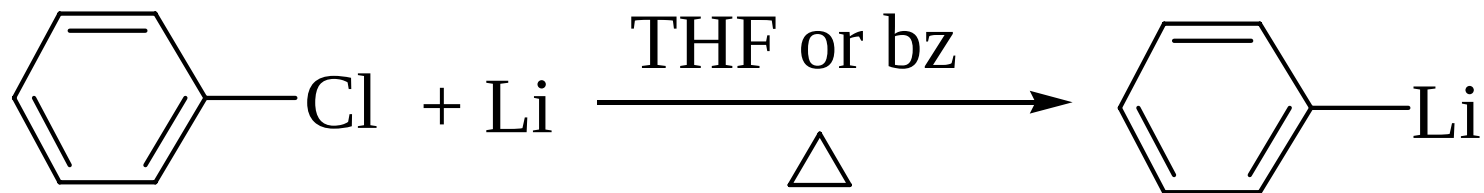
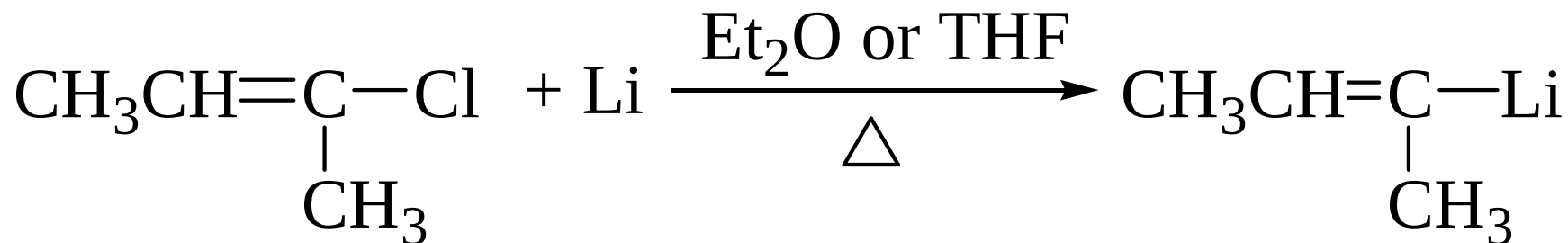
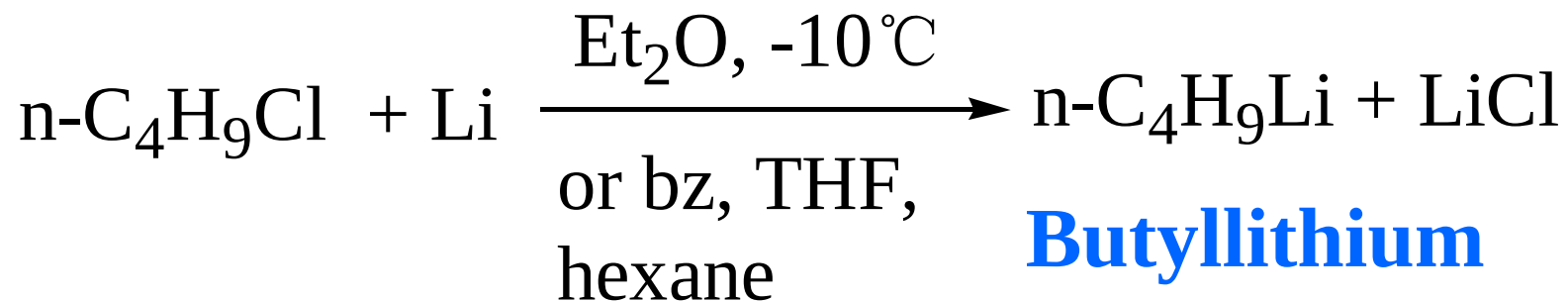
与钠、锂反应



Wurtz 反应

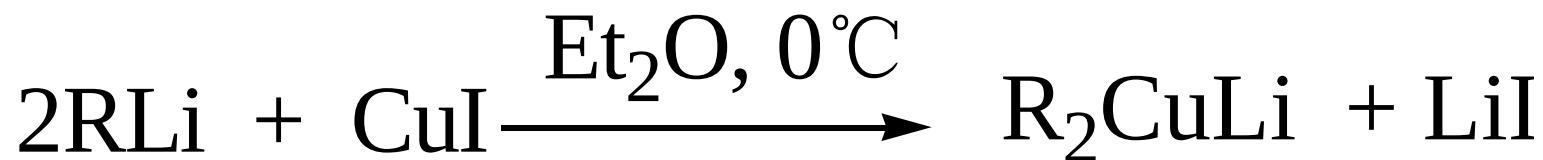


有机锂(烷基锂 Alkylolithiums)

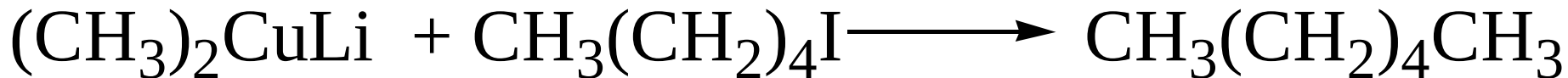


二烷基铜锂(铜锂试剂)

Lithium dialkylcopper



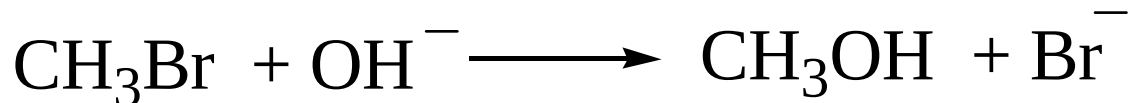
R = 烷基, 烯基, 芳基, 烯丙基



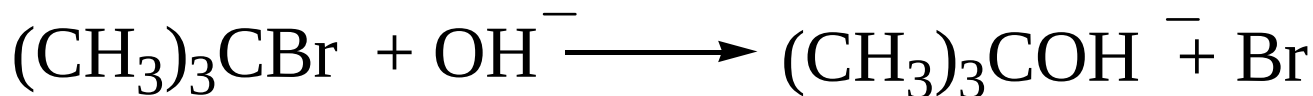
8.4 亲核取代的反应机理



Nucleophilic substitution



$v \propto [\text{CH}_3\text{Br}][\text{OH}^-]$ 二级反应

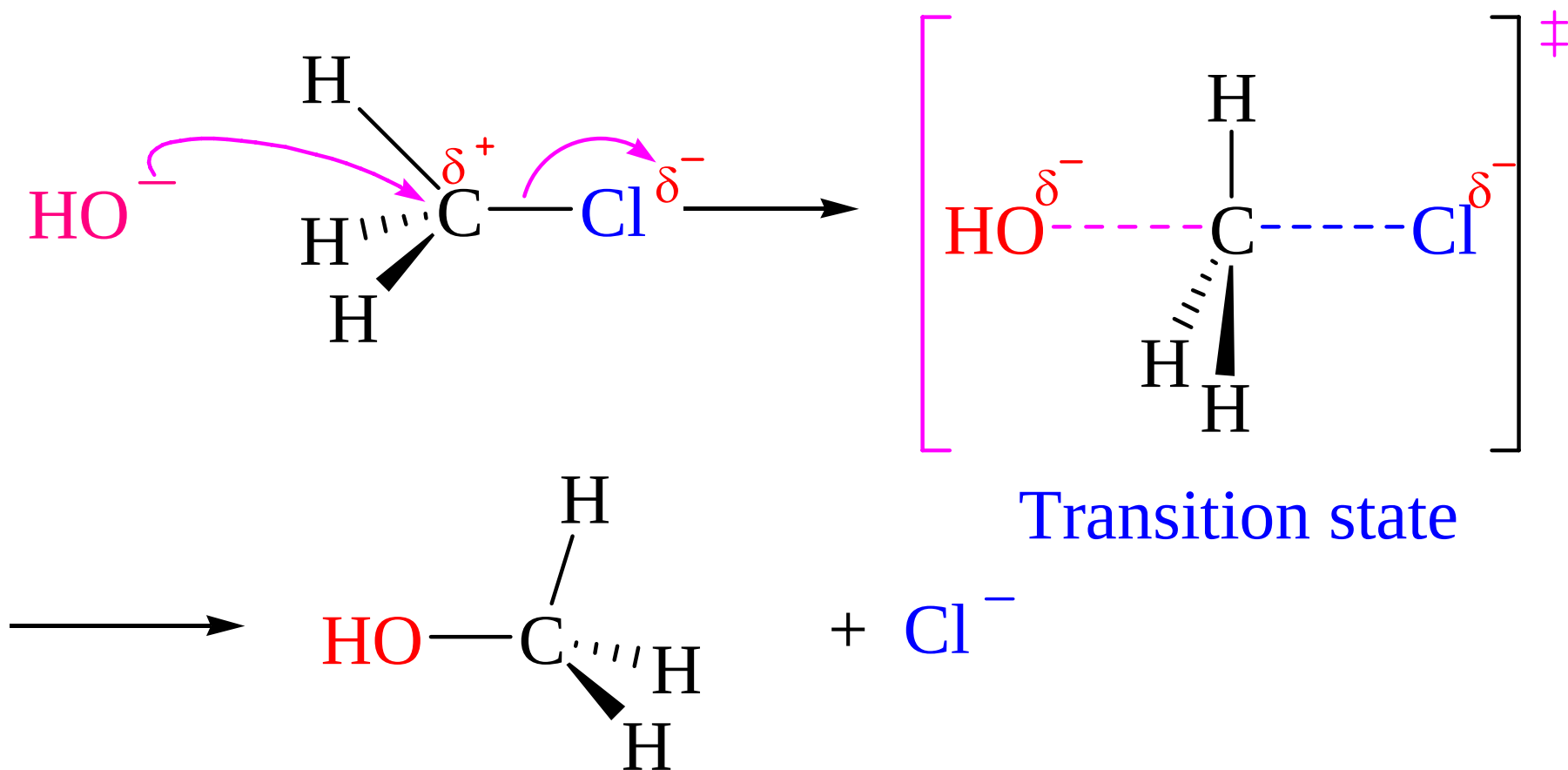


$v \propto [(\text{CH}_3)_3\text{CBr}]$ 一级反应

双分子亲核取代反应($\text{S}_{\text{N}}2$)

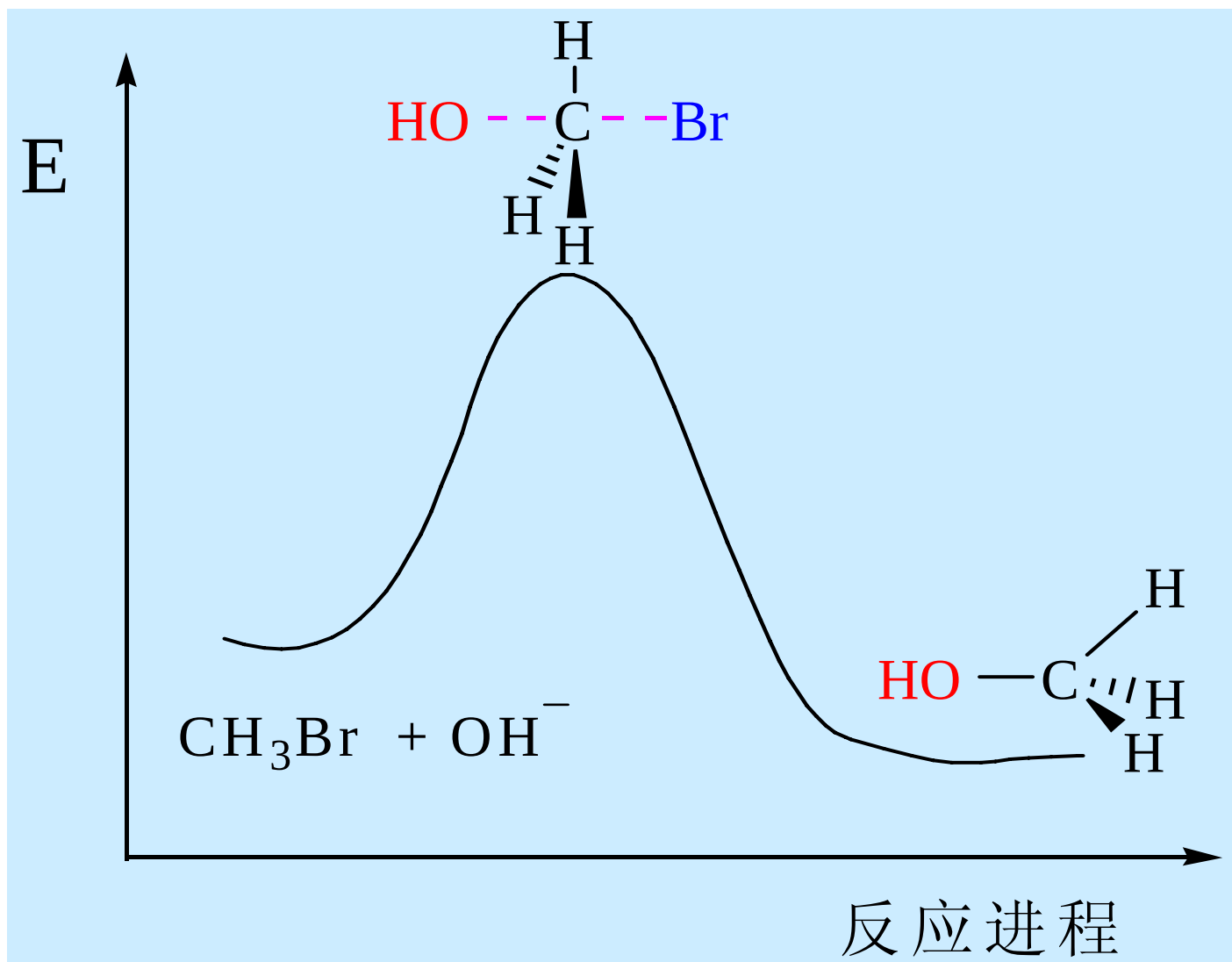
单分子亲核取代反应($\text{S}_{\text{N}}1$)

8.4.1 双分子亲核取代反应(S_N2)



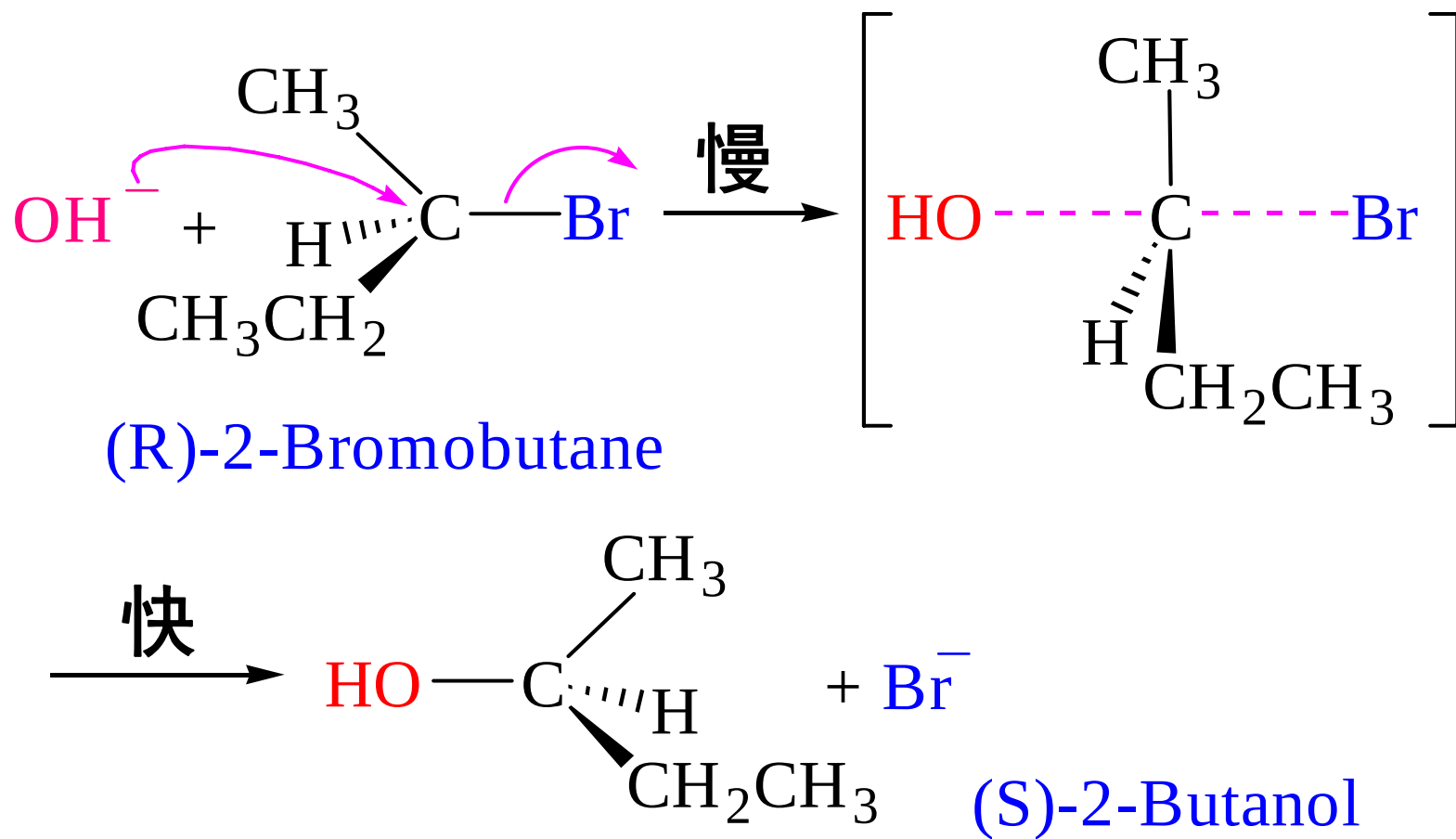
Walden inversion

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构型翻转

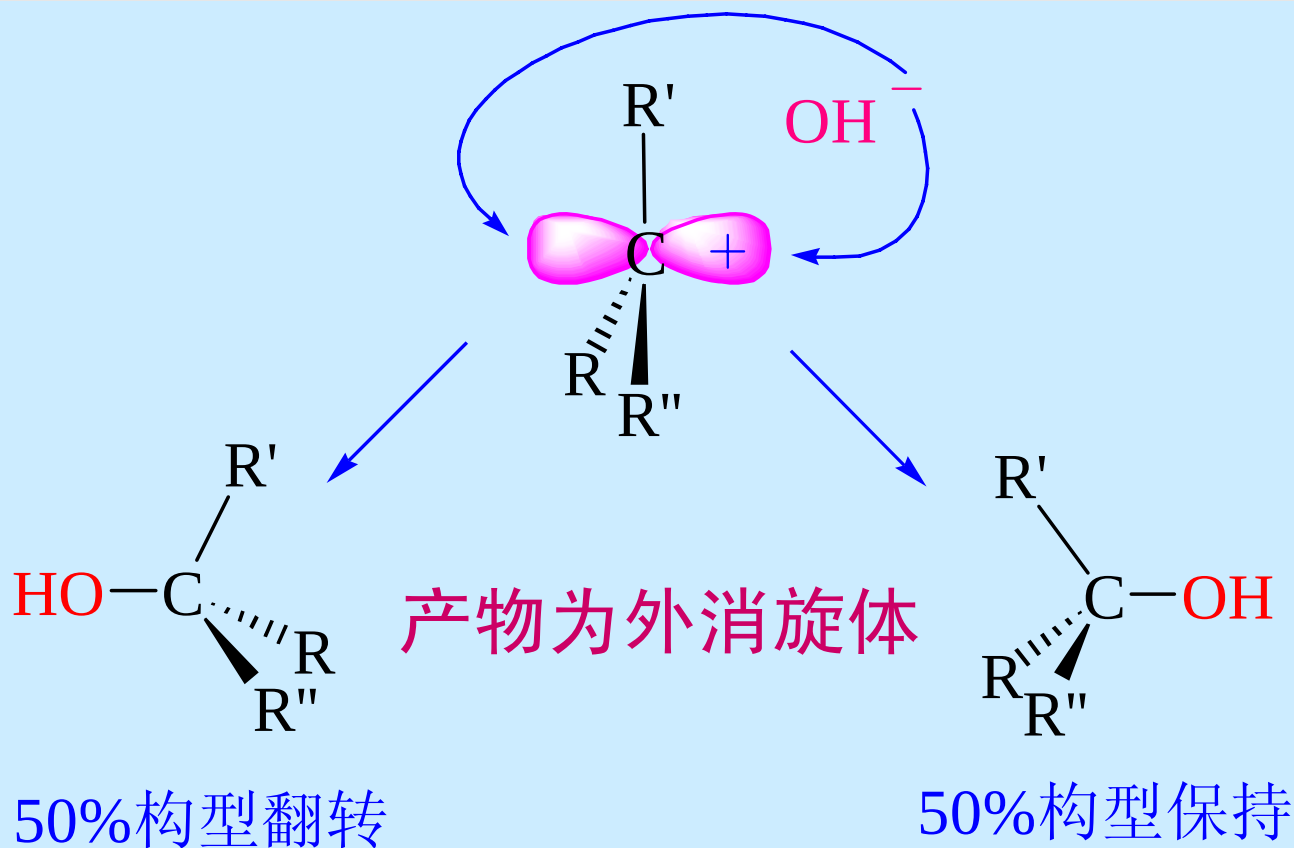
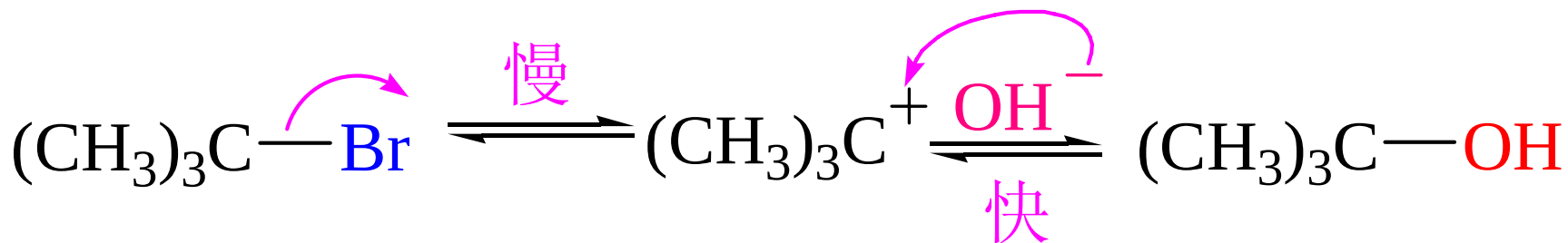


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S_N2 反应特点

- 反应连续进行，旧键断裂与新键形成同时进行
- 构型发生翻转
- 二级反应
- 反应一步完成

8.4.2 单分子亲核取代反应(S_N1)

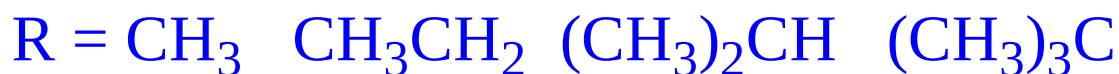


S_N1 反应特点

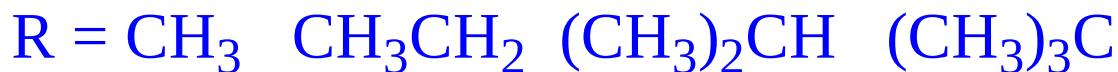
- 分步进行，有活性中间体形成
- 常发生重排
- 反应产物为外消旋体
- 一级反应，反应速率仅与RX浓度有关

8.4.3 影响亲核取代反应的因素

1. 烃基结构



相对速率 1.0 1.7 45 1×10^8



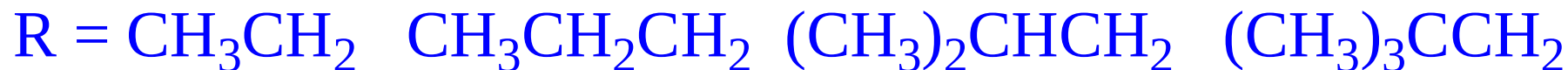
相对速率 150 1.0 0.01 0.001

- $\text{S}_{\text{N}}1$: $3^\circ > 2^\circ > 1^\circ > \text{CH}_3\text{X}$
- $\text{S}_{\text{N}}2$: $\text{CH}_3\text{X} > 1^\circ > 2^\circ > 3^\circ$

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相对速率	150	1.0	0.01	0.001
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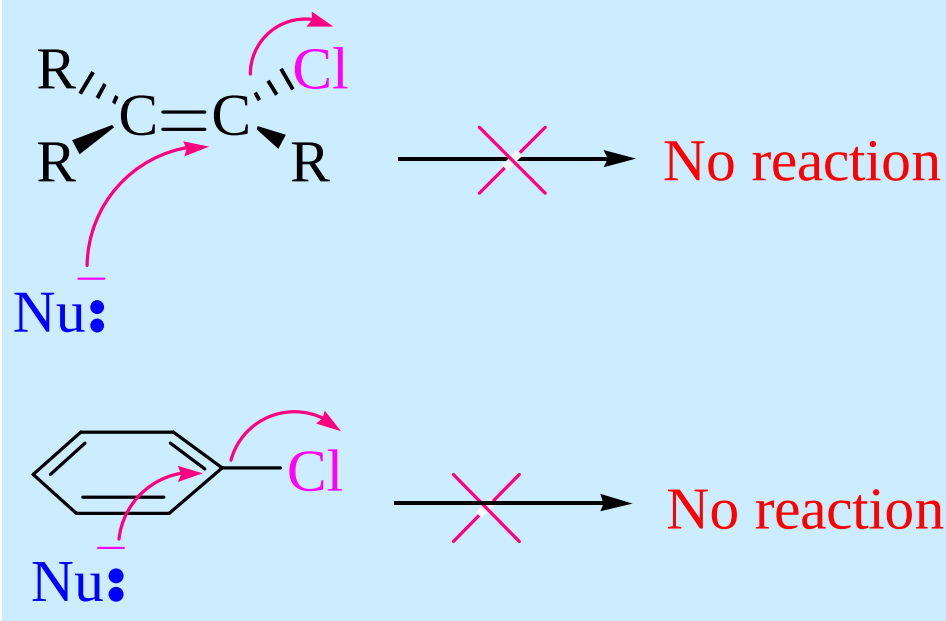


相对速率	1	0.28	0.03	4.2×10^{-6}
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β -H 被烷基取代对反应速率的影响
不如 α -H 被取代明显

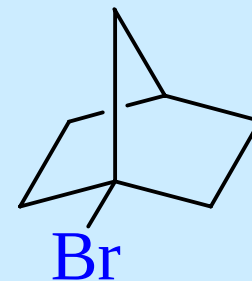
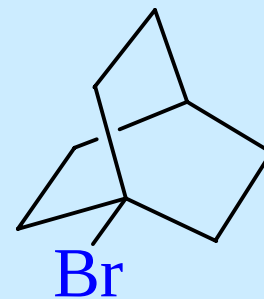
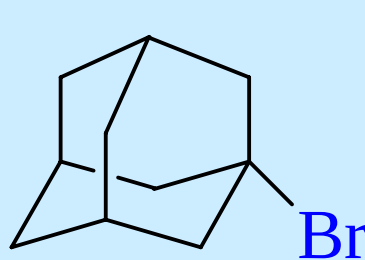
烯丙型 $\text{CH}_2=\text{CHCH}_2\text{X}$ 苄基型 - CH_2X

不论是 $\text{S}_{\text{N}}1$ 还是 $\text{S}_{\text{N}}2$ 反应都很快



$\text{S}_{\text{N}}1$

$(\text{CH}_3)_3\text{CBr}$



Relative rate 1

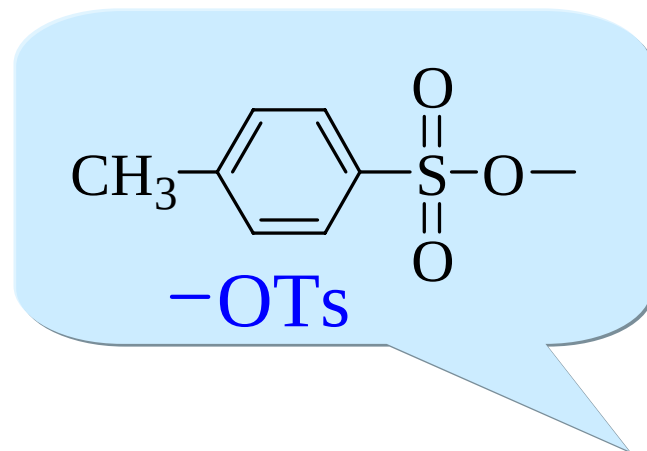
10^{-3}

10^{-6}

10^{-13}

8.4.3 影响亲核取代反应的因素

2. 离去基团



-OH

-OR

-NH₂

-F

-Cl

-Br

-I

-OTs

相对反应性 ~ 0 0.001 0.2 10 30 60



离去能力增强

离去基团的离去能力强，对S_N1和S_N2都有利

8.4.3 影响亲核取代反应的因素

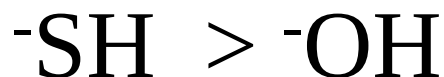
3. 试剂的亲核性

试剂的亲核性强对 S_N2 有利，对 S_N1 影响不大

- 亲核性 —— 试剂与碳原子的结合能力
- 碱性 —— 试剂与质子的结合能力
- 决定试剂亲核性强弱的因素：
 - (1) 试剂的给电子能力
 - (2) 试剂的可极化性

一般情况下亲核性与碱性的顺序是一致的

亲核性



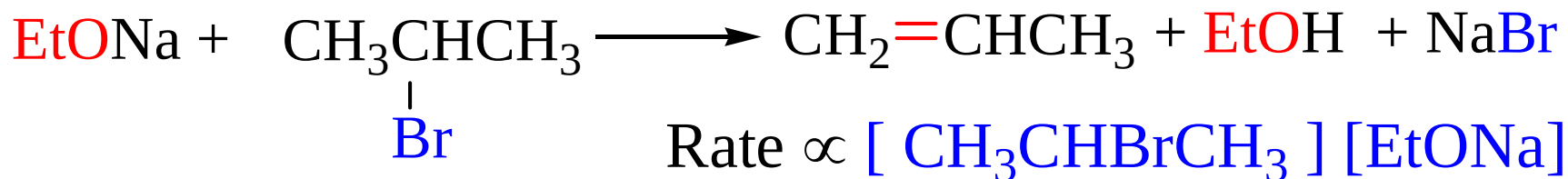
但也有不一致的情况：



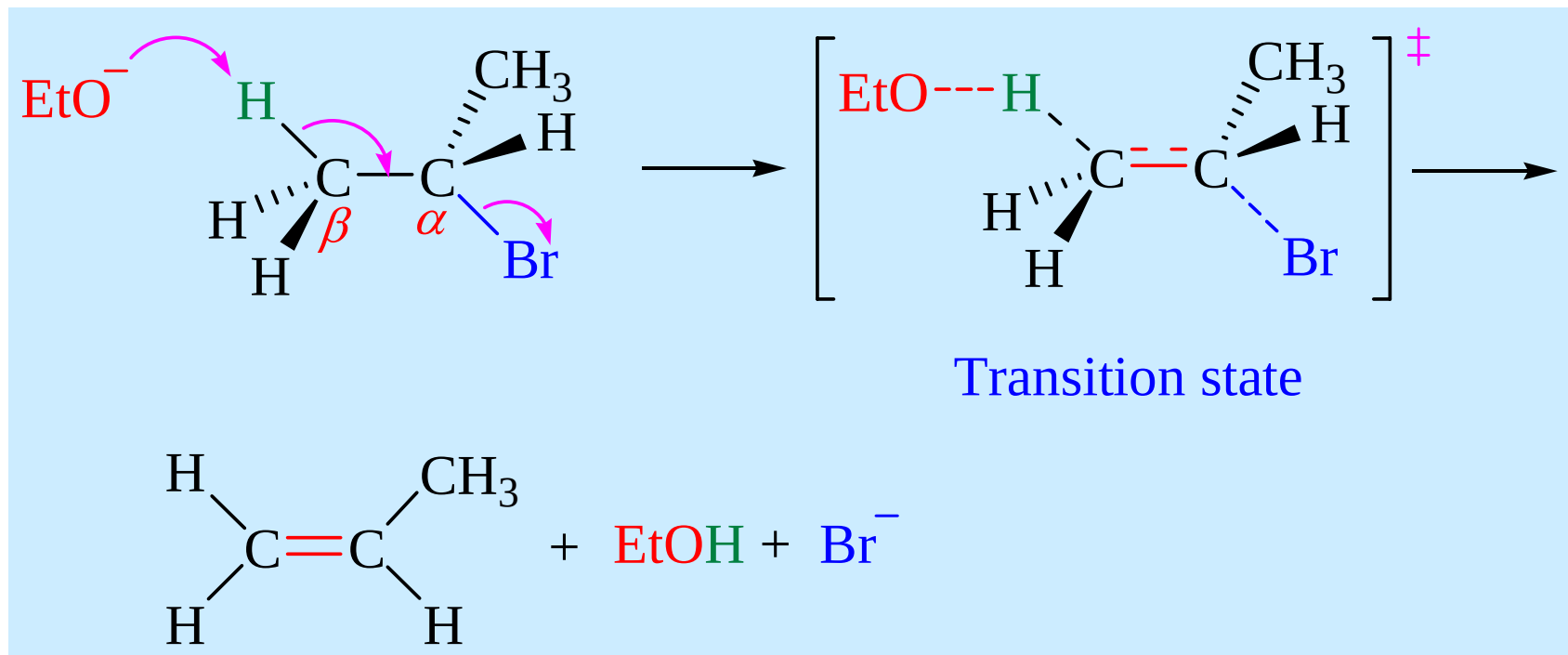
碱性增强，亲核性减弱

8.5 消除反应的反应机理

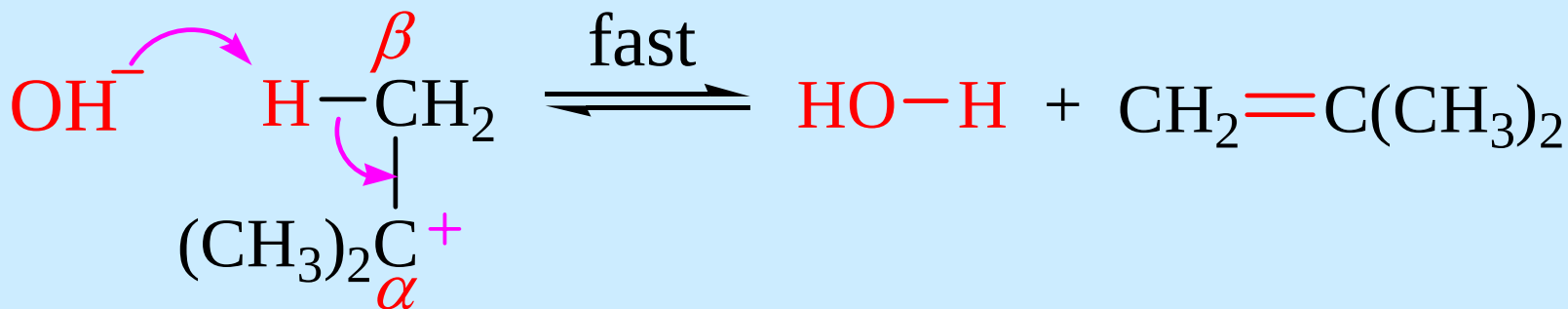
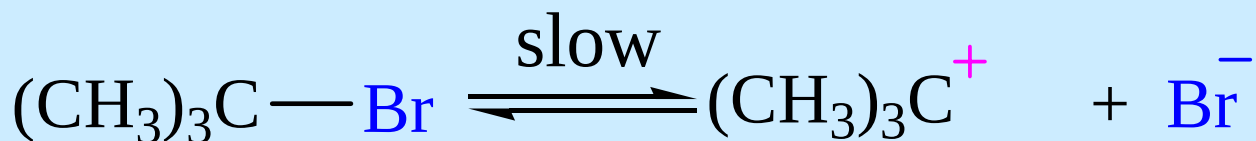
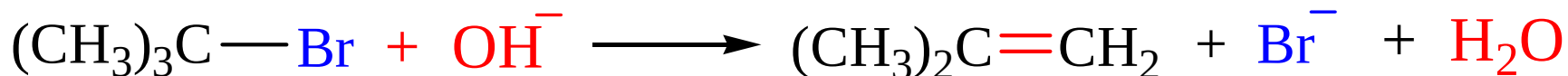
8.5.1 双分子消除反应(E2)



$$\text{Rate } k [\text{CH}_3\text{CHBrCH}_3][\text{EtONa}]$$

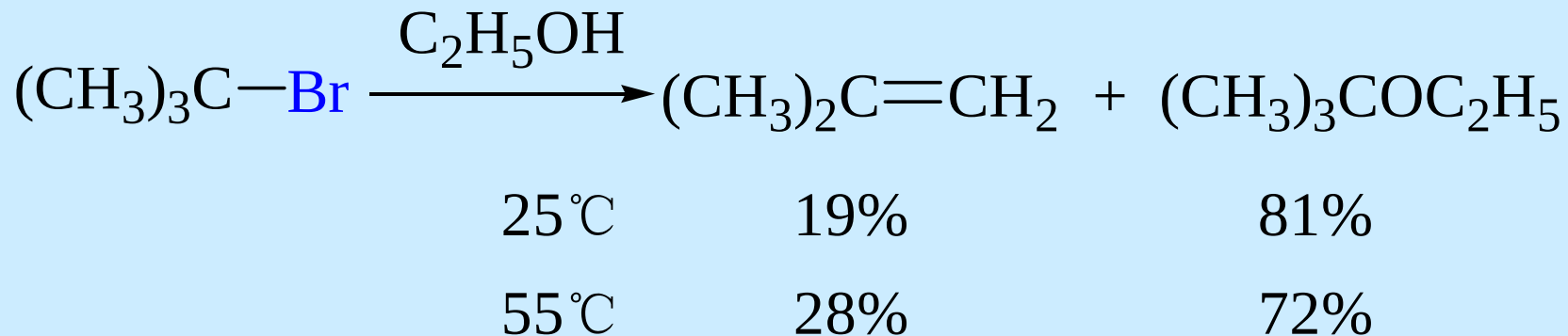


8.5.2 单分子消除反应(E1)

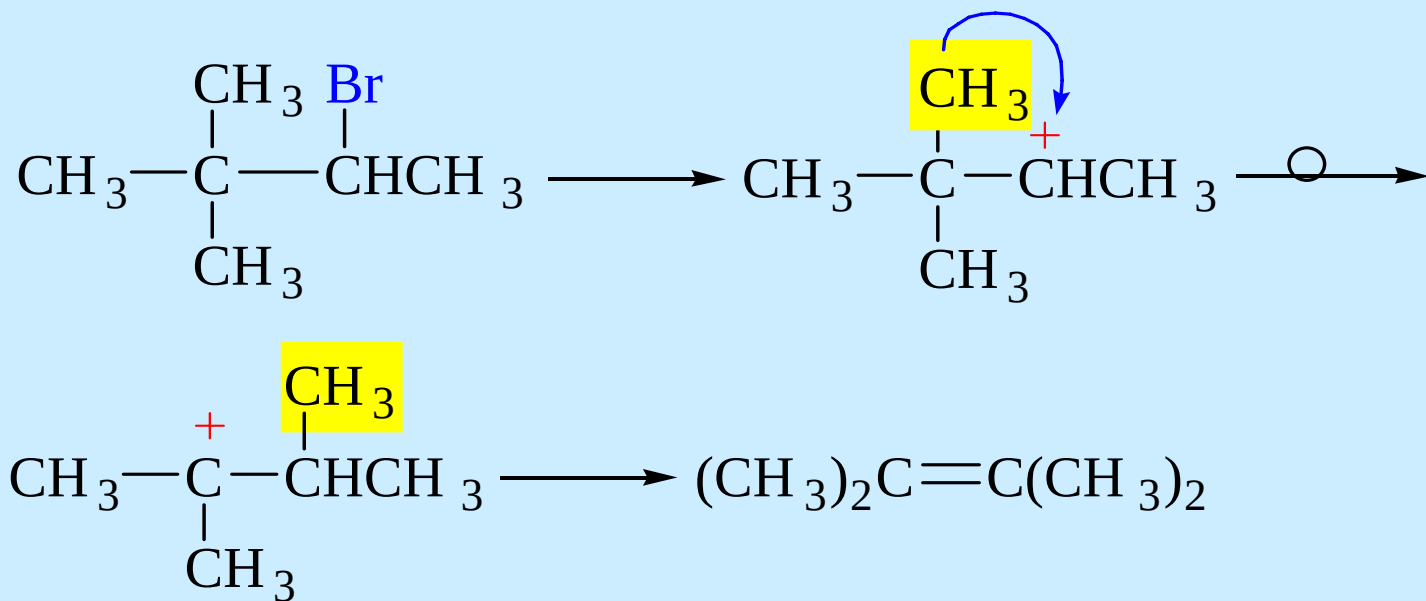


- E1的反应速率只与卤代烃浓度有关
- 发生E1反应不一定要有碱存在，即使有碱存在，碱的浓度和强度对反应速率没影响

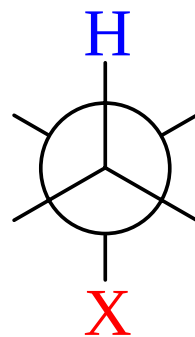
消除反应和亲核取代反应的竞争



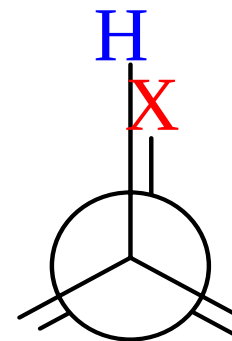
E1可能发生重排



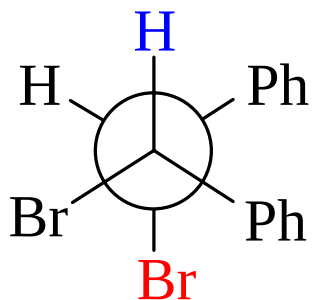
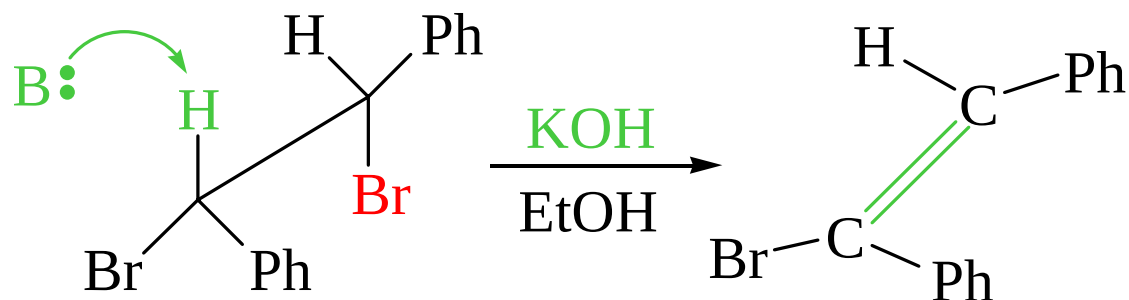
消除反应的立体化学



反式共平面



顺式共平面



主要按反式共平面方式进行消除

8.6 卤代烃的制备

由醇制备



or PX_5

$\text{X} = \text{Br}, \text{Cl}$

其它制备方法自学

- P148: 8.4
- P151: 8.5, 8.6
- P152: 8.7
- 习题: 2.3.4.5.6.7.8.9